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SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Examplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies 25	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues 25	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration 20	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation 10	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results 10	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

$\frac{90}{100}$

Role of the OSI Model in Software-Defined Networking (SDN)

CSA0720-COMPUTER NETWORKS

PRESENTED BY: G MAHESH BABU
SHAIK IMRAN

Introduction

- The OSI (Open Systems Interconnection) Model is a seven-layer networking framework.
- Software-Defined Networking (SDN) separates the control plane from the data plane.
- SDN provides centralized network management and automation.
- The OSI model helps explain how SDN operates across different network layers.
- Together, OSI and SDN improve flexibility, scalability, and efficiency.

Objectives

- Understand the OSI model and its seven layers.
- Explain the concept of SDN.
- Analyze the role of OSI layers in SDN.
- Identify the benefits of OSI with SDN.
- Study network performance improvements.

Problem Statement

- Traditional networks are difficult to manage.
- Manual configuration increases complexity.
- Limited scalability.
- Slow response to failures.
- Lack of centralized control.

Role of the OSI Model in SDN

- Application: Traffic monitoring & security.
- Presentation: Data formatting & encryption.
- Session: Communication sessions.
- Transport: Reliable TCP/UDP delivery.
- Network: Routing via SDN controller.
- Data Link: Switch control (OpenFlow).
- Physical: Hardware transmission.

Applications

- Data centers
- Cloud computing
- Enterprise networks
- ISP networks
- Smart cities
- 5G communication
- Network security

Advantages

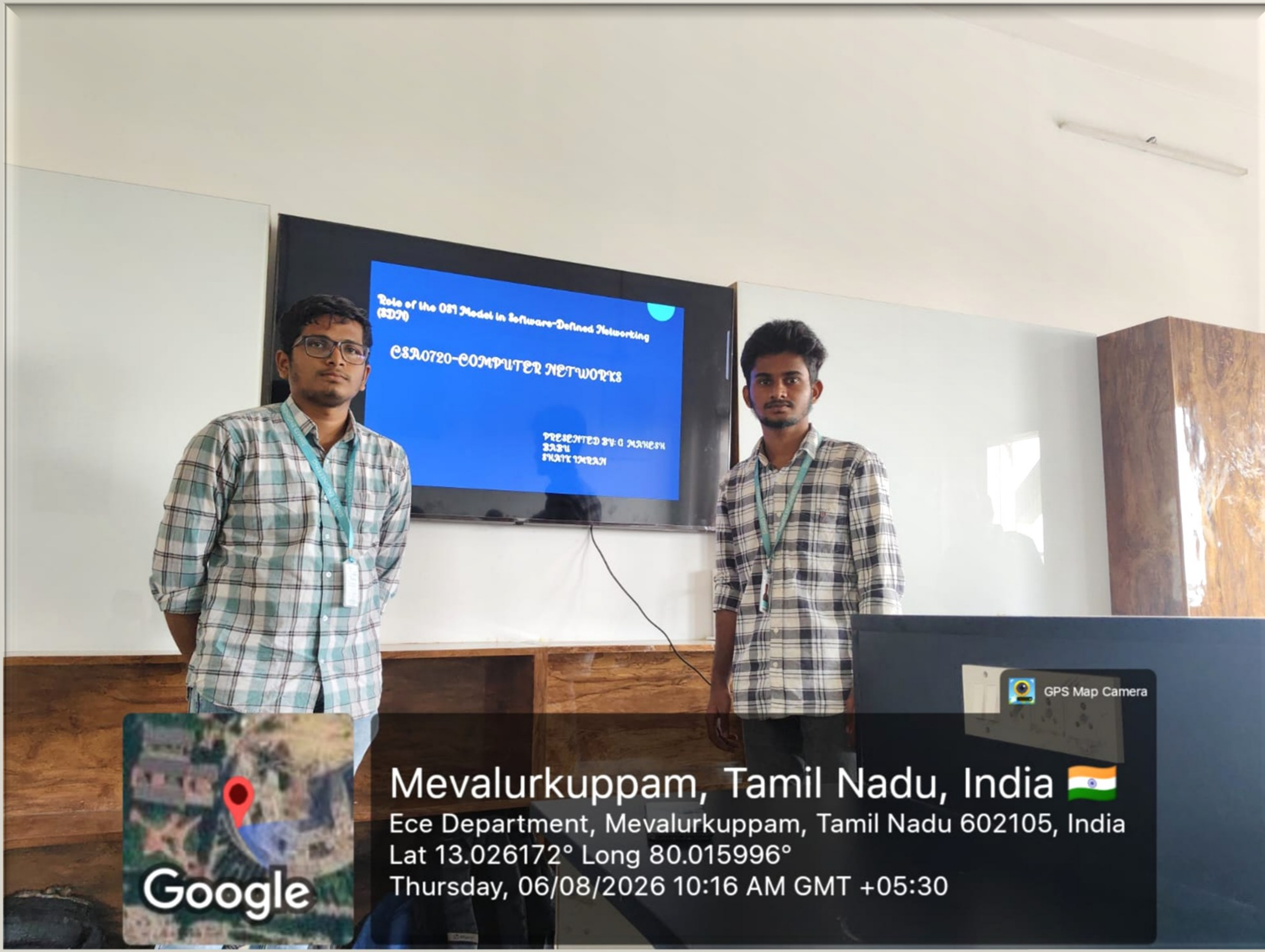
- Centralized control
- Easy management
- High scalability
- Fast configuration
- Better resource use
- Enhanced security
- Lower cost

Conclusion

- OSI provides a structured networking model.
- SDN enables centralized intelligence.
- Improves performance and reliability.
- Supports programmable, scalable networks.

Future Scope

- AI-driven networking
- 5G/6G expansion
- Cloud & edge computing
- Automated cybersecurity
- IoT integration
- Self-healing networks
- Smart industries



Role of the OSI Model in Software-Defined Networking (SDN)
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